KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness for planning tasks

Source: arxiv.org/abs/2504.03553

Goal: regulate cognition + retrieval based on situational need (fast / slow / knowledgeable).

Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts Efficiency

Existing agent learning often applies a 'flood irrigation' strategy.

Injects gold trajectories, external feedback, or domain knowledge regardless of need.

Costs: fragile planning, pattern collapse on unexpected inputs, high inference cost.

Three dominant paradigms (and limitation)

Trajectory imitation (e.g., ReAct): copies behavior; no situational regulation.

Trial-and-error refinement (e.g., Reflexion): reflection everywhere; expensive.

Knowledge-augmented (e.g., KnowAgent/WKM): retrieves everywhere; wasteful.

Human analog: situational self-awareness

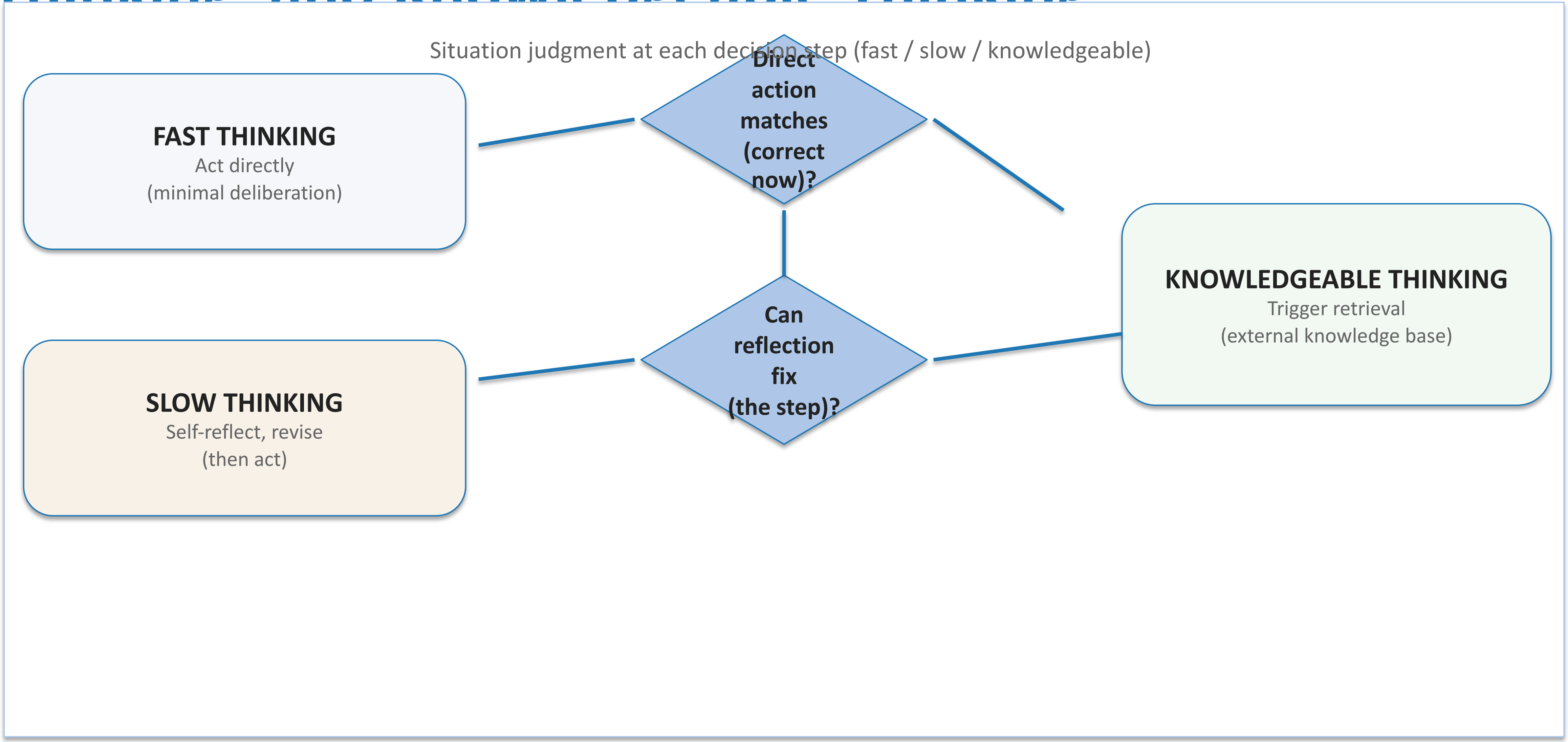
Can I act correctly now?

Do I need to pause and reflect?

Do I need external knowledge/help?

KnowSelf: learn to choose cognition + retrieval only when needed

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf three-situation taxonomy

KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

Pipeline (paper)

1) Data construction: self-explore trajectories; heuristic labels each step as fast/slow/knowledgeable; insert special tokens.

2) Two-stage training: SFT learns token patterns; RPO refines when to emit tokens.

3) Self-aware inference: tokens trigger direct action, reflection, or knowledge retrieval.

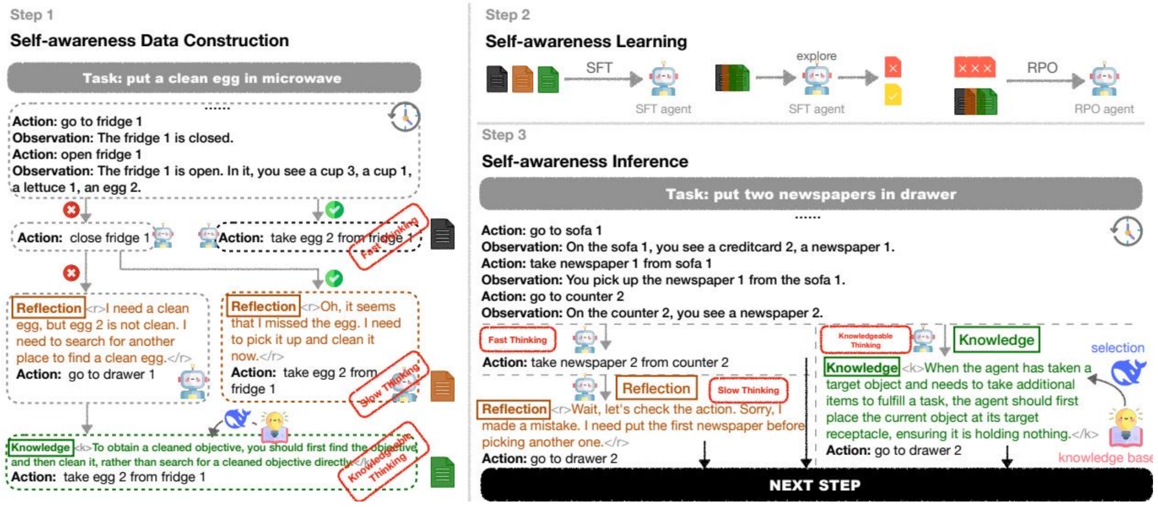
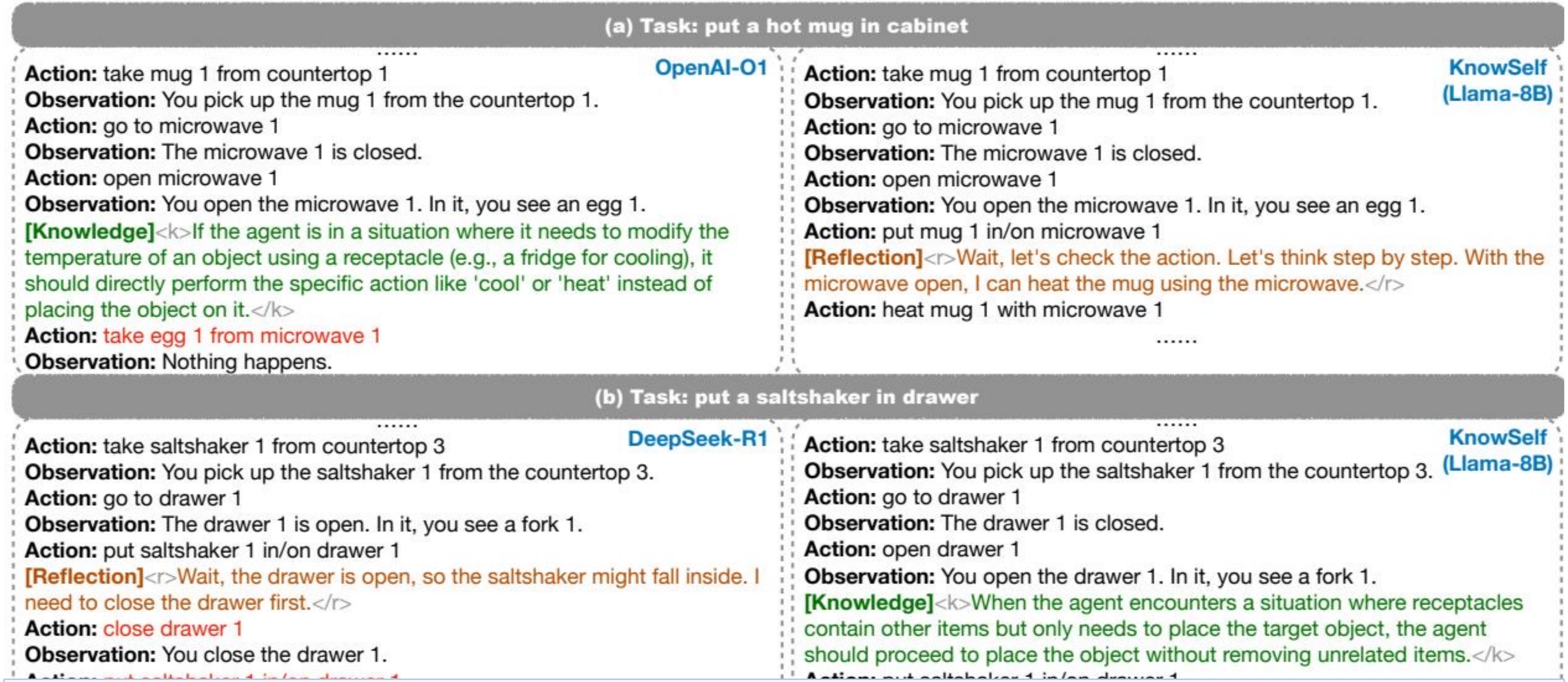


Figure (from source brief): data construction → learning → inference
Method: heuristic labeling + SFT + RPO

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail



Qualitative: reflection helps heat mug correctly; knowledge token helps place saltshaker without irrelevant detours.

Qualitative grounding (Figure 1)

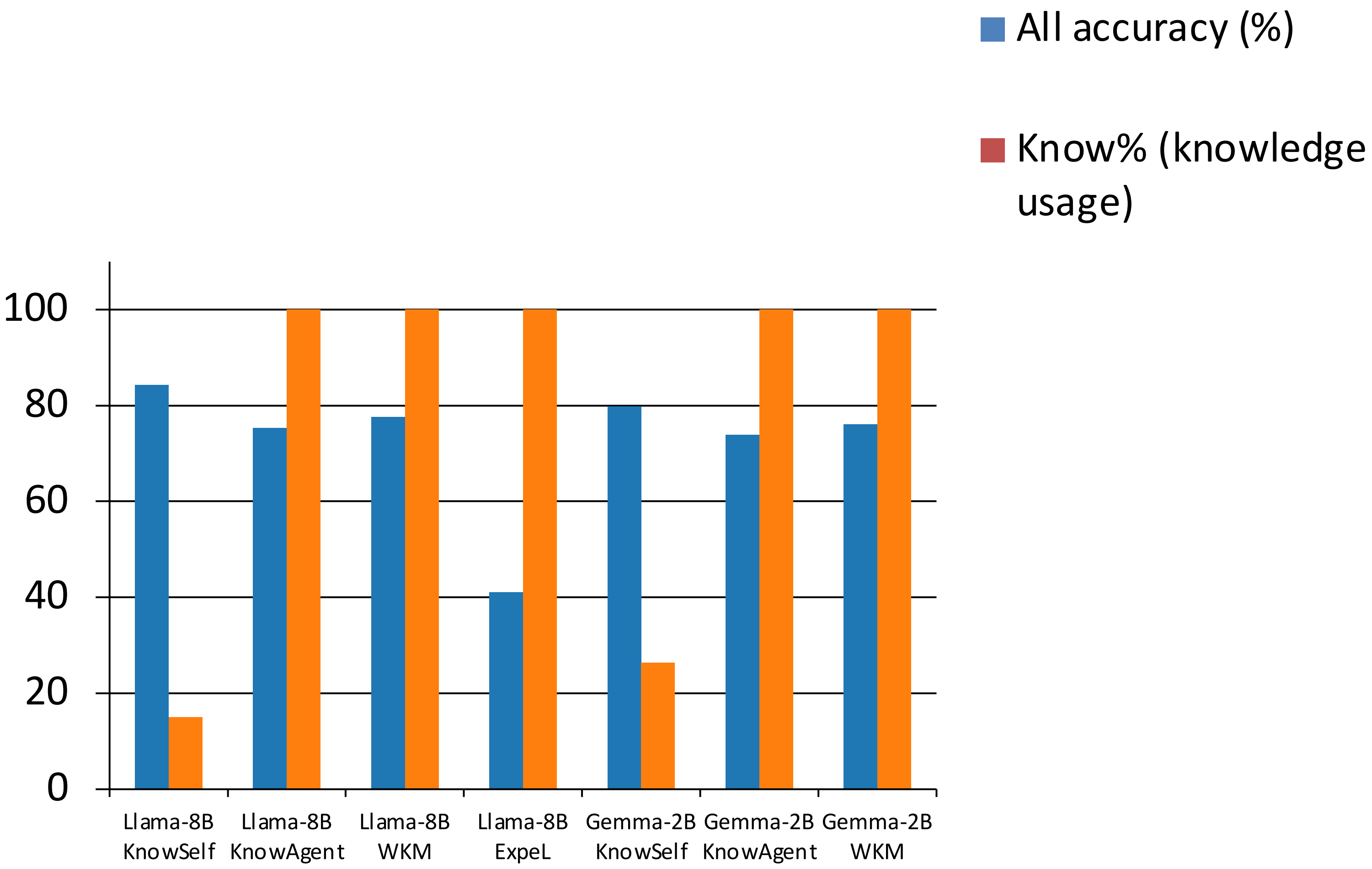
KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	**KnowSelf**	**15.01%**	**91.67**	**87.10**	**91.30**	**85.71**	**77.78**	**64.71**	**84.33**
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12

KnowSelf: 84.33% (Llama-8B) with 15.01% knowledge; 79.85% (Gemma-2B) with 26.41% knowledge.

ALFWorld success rates (from source brief)

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge Methods



Efficiency story

Full-knowledge methods: Know% = 100% (ExpeL / KnowAgent / WKM).

KnowSelf selectively retrieves: 15.01% (Llama-8B), 26.41% (Gemma-2B).

Yet KnowSelf attains top overall success (All).

Scaling: efficiency holds across 2B → 8B backbones.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

Indiscriminate knowledge injection is wasteful and can be suboptimal.

Situational self-awareness is learnable via special tokens + heuristic labeling.

KnowSelf (Llama-8B) beats GPT-4o ReAct: 84.33% vs 76.12% (ALFWorld All).

KnowSelf matches/exceeds 100%-knowledge baselines while using 15–26% knowledge.

Generalizes across model scales (Gemma-2B → Llama-8B).

Future directions

Richer situation criteria beyond action-match heuristics.
Cost-aware policies: optimize retrieval/compute budgets end-to-end.
Broader planning benchmarks and multi-tool environments.